

Unhappy to be Guided: Students' Negative Affect and Digital Nudge Effectiveness in Combating Procrastination Towards Informal Learning

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Abstract

Procrastination has been observed to produce various consequences, yet many students still engage in this behavior. Digital nudging is a promising technique for influencing student procrastination, but its effectiveness varies across individuals' characteristics. Research has neglected digital nudge effectiveness within informal learning environments and the influence of affect, both of which represent promising avenues for deeper investigation. This study investigated the correlations between students' recent positive and negative affect levels and their degrees of procrastination in completing informal learning tasks, following their exposure to digital nudges. Following a diary study which facilitated students' exposure to digital nudges and informal learning tasks, this study collected 88 during-study questionnaire responses from 29 university students, measuring their positive and negative affect levels and degrees of procrastination in completing informal learning tasks. Findings indicated that the students' negative affect levels possessed a significant and moderate positive correlation with their degrees of procrastination. This finding highlighted the potential influence of students' negative affect levels on the digital nudge interventions' effectiveness.

CCS Concepts

• **Human-centered computing** → Human computer interaction (HCI); Empirical studies in HCI; Collaborative and social computing~Collaborative and social computing systems and tools.

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CSCW Companion '25, Bergen, Norway

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ACM ISBN 979-8-4007-1480-1/2025/10

<https://doi.org/10.1145/3715070.3749242>

Keywords

Procrastination, Digital Nudging, Positive and Negative Affect, Informal Learning

ACM Reference Format:

Yong Lin Tam, Simon Junyong Chua, Qi Wen Lim, Kok Khiang Lim, and Chei Sian Lee. 2025. Unhappy to be Guided: Students' Negative Affect and Digital Nudge Effectiveness in Combating Procrastination Towards Informal Learning. In *Companion of the Computer-Supported Cooperative Work and Social Computing (CSCW Companion '25)*, October 18–22, 2025, Bergen, Norway. ACM, New York, NY, USA, 5 pages. <https://doi.org/10.1145/3715070.3749242>

1 Introduction

Procrastination is characterized by an individual's conscious delay in addressing required tasks, despite the ensuing consequences [10], and within an academic context, procrastination can be seen in students' delays in their attending to academic tasks [21]. Students who procrastinate have been observed to experience poorer performance in academic activities [11] and higher degrees of stress and frustration [19]. However, a great proportion of students still engage in procrastination, and a recent study observed that more than half of a sample of North American university students reported themselves as being frequent procrastinators towards university assignment completion [18].

Digital nudging, an intervention technique focused on guiding individuals' behaviors and decision-making processes through the incorporation of specific elements possessing degrees of influencing power into a technology's user interface [30], has demonstrated its promising ability to address this issue through its observed receptiveness within a digital learning environment [12], as well as its previous success in influencing students' procrastination behaviors [2, 4]. The technique utilizes user interface elements that target users' cognitive biases and heuristics [3], and it draws inspiration from the earlier nudge theory [23]. However, digital nudging has been observed to possess a key limitation in which a single nudge intervention may exert a varying degree of influence across individuals with different characteristics [5, 9]. One pair of

factors that exhibits promising relevance in understanding digital nudging effectiveness towards student procrastination is students' positive and negative affect, or emotional states [29], which have been observed to be associated with student procrastination [7, 17, 24].

Moreover, previous attempts to influence students' procrastination behaviors with digital nudging have neglected the context of informal learning [2, 4, 16], which encompasses opportunities for learning that exist outside of a traditional classroom [6]. One medium through which informal learning may be facilitated and further investigated is individuals' collaboration with a generative AI chatbot, in which messages may be exchanged between individuals and the chatbot to facilitate learning, thereby exemplifying human–AI collaboration [27]. Hence, this study investigates the relationship between students' positive and negative affect levels and their degrees of procrastination in completing informal learning tasks involving support from a generative AI chatbot, following their exposure to digital nudges. More specifically, it seeks to answer the following research question: How do students' recent positive and negative affect levels correlate with their degrees of procrastination in completing an informal learning task, following their exposure to digital nudge interventions?

Through a diary study involving the facilitation of digital nudge interventions and informal learning tasks, university students' positive and negative affect levels within the past 24 hours were measured and analyzed alongside their degrees of procrastination in completing informal learning tasks, following their exposure to the digital nudge interventions, to identify potential correlations. The study contributes to existing research concerning individuals' characteristics and digital nudge intervention effectiveness, and its findings inspire practical design considerations concerning the effective development of digital nudge interventions targeting student procrastination towards informal learning. Moreover, the study contributes to CSCW research through its employment of a generative AI chatbot-based informal learning web application that enables informal learning to be achieved through human–AI collaboration as opposed to human–human collaboration, therefore inspiring further investigation into the implications arising from human–AI collaboration.

2 Related Work

2.1 Digital Nudging and Student Procrastination

Past studies have successfully implemented digital nudge interventions for inspiring behavioral changes across various contexts, such as improving academic submission rates, guiding the selection of food recipes, and influencing energy savings behavior [8, 15, 25]. Researchers' attempts to influence student procrastination behaviors have also encountered similar success [2, 4]. For instance, nudges involving task commitments and the public display of students' committed tasks on an online platform were observed to inspire reductions in students' procrastination behaviors towards submitting assignments [4], while nudges involving the provision of useful information for completing class activities through a virtual learning environment, which reduced the required effort to

engage in the activities, were observed to inspire reduced procrastination behaviors from students who interacted with the nudges [2].

Conversely, nudges have also been observed to have limited influence on students' procrastination behaviors [16]. More specifically, nudges involving the provision of information comparing students' assignment submission timings, as well as the associated differences in grade performances, were found to have no impact on students' submission timings [16]. However, a significant gap remains in understanding how affect can inform the design of emotionally intelligent and personalized nudges that align with students' motivational states, thereby enhancing the effectiveness of interventions aimed at reducing procrastination. This study aims to fill this gap.

2.2 Students' Positive and Negative Affect and Their Procrastination Levels

Previous studies have also identified the existence of relationships between students' positive and negative affect and their procrastination levels [7, 17, 24]. In a diary study involving 53 undergraduate students, it was observed that the students' negative affect levels possessed a significant positive correlation with their procrastination levels experienced on the following day, while students' positive affect levels were observed to possess a negative correlation with their procrastination levels [17]. Concerning the variables' directional influence, it was found that the students' negative affect levels served to be predictors of their procrastination levels on the following day [17]. Moreover, a separate study observed the existence of a significant positive correlation between students' negative affect levels and their procrastination levels through an analysis of survey questionnaire responses from university students in China [24]. Similarly, another study involving survey questionnaire responses from undergraduate students in the United States supported the presence of a significant positive correlation between students' negative affect levels and their procrastination levels and also identified the presence of a significant negative correlation between students' positive affect levels and procrastination levels [7].

However, one study observed that students' positive affect levels failed to have a directional influence on their procrastination levels, identifying that the students' positive affect levels did not serve as predictors of their procrastination levels on the following day [17]. Nevertheless, researchers' observations concerning the existence of relationships between students' positive and negative affect and their procrastination levels inspire the investigation of correlations between students' recent positive and negative affect levels and the effectiveness of digital nudge interventions in influencing the students' degrees of procrastination.

3 Method

3.1 Participants and Procedure

A 14-day diary study facilitated the digital collection of during-study questionnaire responses from 29 university students recruited from a single Singapore university. The diary study was approved by the researchers' university institutional review board (IRB-2024-561). Participant recruitment was conducted through convenience

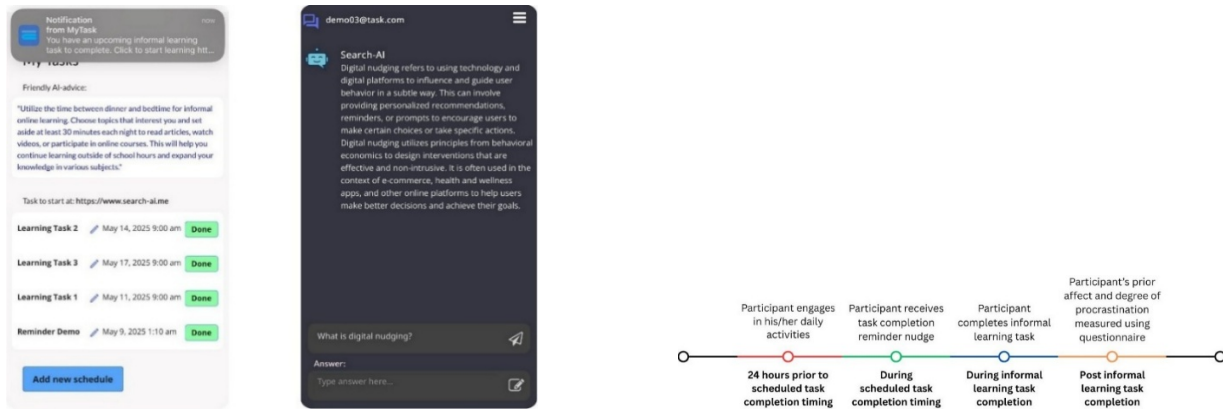


Figure 1: Digital nudging and informal learning web applications (left); diary study during-study procedure (right).

sampling, and the participant inclusion criteria included individuals aged 18 to 65 years old and who were enrolled in the university during the conducting of the study. Concerning participants' demographics, 24.1% were male and 75.9% were female, 6.9% were undergraduate students and 93.1% were graduate students, and 37.9% were STEMM (science, technology, engineering, math, and medicine) students and 62.1% were non-STEMM students. Moreover, participants who completed the diary study were compensated with SGD 50 for their time and effort.

The diary study involved participants' interaction with a digital nudging web application, as well as an informal learning web application (see Figure 1, left). The digital nudging web application facilitated the digital nudge interventions employed within the study, for use with the participants' mobile smartphones. The interventions included scheduled reminders that were delivered through mobile push notifications to serve as timing-based prompts, as well as automatically generated informal learning task completion schedules that served as default option-based nudges and which could be freely modified by participants. On the other hand, the study's informal learning web application provided a simulated environment that facilitated the completion of informal learning tasks through participants' collaboration with a generative AI chatbot. The applications presented in Figure 1 (left) were developed as part of a larger research project.

To commence the diary study, participants were required to perform a first-time sign-in into the digital nudging web application, which then automatically generated an informal learning task completion schedule that consisted of three informal learning tasks to be completed at scheduled timings throughout a 14-day duration. Over the course of the diary study, participants received task completion reminders that were delivered from the digital nudging web application, which were sent at timings indicated within the task completion schedule. Upon receipt of a reminder, participants were then required to access the informal learning web application to complete their informal learning task, and upon completion, participants were requested to submit a during-study questionnaire. This procedure is illustrated in Figure 1 (right).

3.2 Materials and Statistical Analyses

The during-study questionnaire measured participants' degrees of procrastination using the Procrastination Assessment Scale for Students (PASS) [20], and it also measured participants' positive and negative affect levels in the past 24 hours using the Positive and Negative Affect Schedule (PANAS) [28]. More specifically, the first section of the PASS, which measures students' degrees of procrastination in various academic activities, was used in the study within the context of informal learning task completion, and it comprised the 3 items of "To what degree do you procrastinate on this task?", "To what degree is procrastination on this task a problem for you?", and "To what extent do you want to decrease your tendency to procrastinate on this task?". Moreover, the items were assessed using a 5-point Likert scale, in which a higher value demonstrated a stronger degree of procrastination. On the other hand, the PANAS contained 20 items, where each item measured a different affect, such as distress and excitement, and the items measuring positive and negative affect were then summed separately to attain the final respective scores. Additionally, the items were responded to using a 5-point Likert scale, in which a higher value demonstrated a higher perceived experience of the affect.

Analyses performed on the data included measurement reliability analyses, as well as repeated measures correlation analyses [1], which were performed using IBM SPSS Statistics (Version 30.0) and Python (Version 3.12.3). Furthermore, the open-source Python library pandas (Version 2.2.2) [14] and the open-source Python package Pingouin (Version 0.5.5) [26] were used alongside Python. Moreover, the analyses were conducted using a significance level of 0.05.

4 Results

A Cronbach's alpha value (α) was attained for each measure, and the measures for positive affect ($\alpha = .92$), negative affect ($\alpha = .89$), and degree of procrastination ($\alpha = .88$) were deemed to possess acceptable internal consistency reliability [22].

Following, repeated measures correlation (r_{rm}) analyses were conducted between participants' positive and negative affect levels and their degrees of procrastination to identify the existence of

Table 1: Repeated measures correlation analyses between positive and negative affect and degree of procrastination

	1.	2.	3.
1. Positive Affect	1		
2. Negative Affect	.26*	1	
3. Degree of Procrastination	-.02	.42**	1

Note: * $p < .05$, ** $p < .01$.

Table 2: Repeated measures correlation analyses between individual negative affects and degree of procrastination

	Dis.	Ups.	Gui.	Sca.	Hos.	Irr.	Ash.	Ner.	Jit.	Afr.
Degree of Procrastination	.42**	.41**	.39**	.33*	-.04	.22	.48**	.34**	.16	.34**

Note: Dis. = Distressed, Ups. = Upset, Gui. = Guilty, Sca. = Scared, Hos. = Hostile, Irr. = Irritable, Ash. = Ashamed, Ner. = Nervous, Jit. = Jittery, Afr. = Afraid; * $p < .05$, ** $p < .01$.

significant correlations. A total of 88 responses from 29 participants were analyzed, and Table 1 details the results from the repeated measures correlation analyses.

From Table 1, it can be observed that positive affect possessed a non-significant negative correlation with degree of procrastination, $r_{rm}(58) = -.02$, $p < .89$. Moreover, positive affect was also observed to possess a significant positive correlation with negative affect, $r_{rm}(58) = .26$, $p < .046$. The first finding indicated that participants' recent positive affect levels within the past 24 hours possessed a weak negative correlation with their degrees of procrastination in completing the informal learning tasks, following their exposure to the digital nudge interventions, but the correlation was not found to be statistically significant.

On the other hand, it can also be observed from Table 1 that negative affect exhibited a significant positive correlation with degree of procrastination, $r_{rm}(58) = .42$, $p < .001$. This indicated that participants' recent negative affect levels within the past 24 hours possessed a moderate positive correlation with their degrees of procrastination in completing the informal learning tasks, following their exposure to the digital nudge interventions, and the correlation was also found to be statistically significant. Additionally, when further investigating the individual affects that comprise negative affect (see Table 2), it was observed that ashamed ($r_{rm} = .48$, $p < .001$), distressed ($r_{rm} = .42$, $p < .001$), upset ($r_{rm} = .41$, $p = .001$), and guilty ($r_{rm} = .39$, $p = .002$) possessed the strongest significant correlations with degree of procrastination.

5 Discussion and Conclusion

From the study's findings, it was discovered that the students' negative affect levels possessed a significant and moderate positive correlation with their degrees of procrastination. Conversely, it was observed that the students' positive affect levels possessed a non-significant and weak negative correlation with their degrees of procrastination. However, despite the non-significance of the correlation between positive affect and degree of procrastination, the correlation's direction was found to be consistent with previous studies' findings, where positive affect was observed to possess a negative correlation with student procrastination [7, 17]. Additionally, the presence of a significant positive correlation between

negative affect and degree of procrastination was found to be consistent with previous studies' findings as well [7, 17, 24].

Most importantly, the key identification of a significant and moderate positive correlation between students' recent negative affect levels and their degrees of procrastination, following their exposure to the digital nudge interventions, highlights the potential importance of negative affect in influencing the interventions' effectiveness. This may inspire subsequent investigations that examine the predictive power of negative affect and its individual affects in influencing changes in the interventions' effectiveness. Moreover, studies may investigate the use of digital nudges themselves to influence users' affect, such as the use of displays of natural landscapes [13].

Furthermore, future studies may examine the impact of the medium used for facilitating informal learning on students' degrees of procrastination by measuring and comparing results involving informal learning tasks facilitated through human-AI collaboration and human-human collaboration mediums.

Separately, the study's limitations included generalizability, sampling bias, and measurement accuracy. Specifically, the sample size of 29 students may limit the findings' generalizability. Moreover, the use of convenience sampling resulted in unequal proportions of participant demographics that may also affect the study's generalizability, such as the unequal proportion of undergraduate and graduate students, as participants' varying susceptibilities may affect the observed effectiveness of the digital nudge interventions. In addition, the use of survey questionnaires as measurement instruments may affect the measurements' accuracy, as the measurements were self-reported and based on participants' perceptions, therefore reducing their objectivity.

All in all, the study investigated the relationship between students' recent positive and negative affect levels and their degrees of procrastination in completing informal learning tasks, following their exposure to digital nudge interventions. Additionally, the study explored informal learning facilitated through human-AI collaboration and inspires further investigation into the implications of human-AI collaboration within the context of informal learning and student procrastination. Importantly, findings indicated that the students' negative affect levels possessed a significant and

moderate positive correlation with their degrees of procrastination. This preliminary finding contributes to existing research surrounding the impact of individuals' characteristics on the effectiveness and sustainability of digital nudge interventions. Specifically, it highlights students' negative emotional states as being a potentially important practical design consideration for the development of effective digital nudge interventions targeting student procrastination towards informal learning.

Acknowledgments

This research is supported by the Ministry of Education, Singapore, under its Tertiary Education Research Fund (Award MOE2023-TRF-028). Any opinions, findings, conclusions, or recommendations expressed in this material are those of the author(s) and do not reflect the views of the Ministry of Education, Singapore.

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